

# Language Transfer

**Cross-linguistic influence  
in language learning**

**Terence Odlin**

---

C A M B R I D G E  
Applied Linguistics

---

Series Editors: Michael H. Long and Jack C. Richards

# Language Transfer

**Cross-linguistic influence  
in language learning**

**Terence Odlin**

---

C A M B R I D G E  
Applied Linguistics

---

Series Editors: Michael H. Long and Jack C. Richards

## Language Transfer

## THE CAMBRIDGE APPLIED LINGUISTICS SERIES

Series editors: Michael H. Long and Jack C. Richards

This new series presents the findings of recent work in applied linguistics which are of direct relevance to language teaching and learning and of particular interest to applied linguists, researchers, language teachers, and teacher trainers.

*In this series:*

**Interactive Approaches to Second Language Reading** *edited by Patricia L. Carrell, Joanne Devine, and David E. Eskey*

**Second Language Classrooms – Research on teaching and learning** *by Craig Chaudron*

**Language Learning and Deafness** *edited by Michael Strong*

**The Learner-Centred Curriculum** *by David Nunan*

**The Second Language Curriculum** *edited by Robert Keith Johnson*

**Language Transfer – Cross-linguistic influence in language learning** *by Terence Odlin*

**Linguistic Perspectives on Second Language Acquisition** *edited by Susan M. Gass and Jaquelyn Schachter*

**Learning Strategies in Second Language Acquisition** *by J. Michael O'Malley and Anna Uhl Chamot*

**The Development of Second Language Proficiency** *edited by Birgit Harley, Patrick Allen, Jim Cummins and Merrill Swain*

**Second Language Writing – Research insights for the classroom** *edited by Barbara Kroll*

**Genre Analysis – English in academic and research settings** *by John M. Swales*

**Evaluating Second Language Education** *edited by J. Charles Alderson and Alan Beretta*

**Perspectives on Pedagogical Grammar** *edited by Terence Odlin*

**Academic Listening** *edited by John Flowerdew*

**Power and Inequality in Language Education** *edited by James W. Tollefson*

**Language Program Evaluation – Theory and practice** *by Brian K. Lynch*

**Sociolinguistics and Language Teaching** *edited by Sandra Lee McKay and Nancy H. Hornberger*

**Contrastive Rhetoric – Cross-cultural aspects of second language writing** *by Ulla Connor*

**Teacher Cognition in Language Teaching – Beliefs, decision-making, and classroom practice** *by Devon Woods*

**Second Language Vocabulary Acquisition – A rationale for pedagogy** *edited by James Coady and Thomas Huckin*

# Language Transfer

Cross-linguistic influence in  
language learning

*Terence Odlin*

The Ohio State University



**CAMBRIDGE**  
UNIVERSITY PRESS

PUBLISHED BY THE PRESS SYNDICATE OF THE UNIVERSITY OF CAMBRIDGE  
The Pitt Building, Trumpington Street, Cambridge CB2 1RP, United Kingdom

CAMBRIDGE UNIVERSITY PRESS  
The Edinburgh Building, Cambridge CB2 2RU, United Kingdom  
40 West 20th Street, New York, NY 10011-4211, USA  
10 Stamford Road, Oakleigh, Melbourne 3166, Australia

© Cambridge University Press 1989

This book is in copyright. Subject to statutory exception  
and to the provisions of relevant collective licensing agreements,  
no reproduction of any part may take place without  
the written permission of Cambridge University Press.

First published 1989  
Sixth printing 1997

Typeset in Sabon

*Library of Congress Cataloging-in-Publication Data*

Odlin, Terence.

Language transfer / Terence Odlin.

p. cm. – (Cambridge applied linguistics series)

Bibliography: p.

Includes index.

ISBN 0-521-37168-6 hardback. ISBN 0-521-37809-5 paperback

1. Language transfer (Language learning). I. Title. II. Series.

P118.25.035 1989

418 – dc19

88-30760

CIP

*British Library Cataloging in Publication Data*

Odlin, Terence

Language transfer. – (The Cambridge applied  
linguistics series).

1. Foreign languages. Learning. Role of  
native languages.

I. Title

418'.007

ISBN 0-521-37168-6 hardback

ISBN 0-521-37809-5 paperback

Transferred to digital printing 2003

In memory of Walter Odlin  
1908–1985





# *Contents*

|                                                             |            |
|-------------------------------------------------------------|------------|
| Series editors' preface                                     | ix         |
| Preface                                                     | xi         |
| <b>1 Introduction</b>                                       | <b>1</b>   |
| <b>2 Earlier thinking on transfer</b>                       | <b>6</b>   |
| 2.1 Languages (and dialects) in contact                     | 6          |
| 2.2 Transfer as a controversy in language teaching          | 15         |
| <b>3 Some fundamental problems in the study of transfer</b> | <b>25</b>  |
| 3.1 Problems of definition                                  | 25         |
| 3.2 Problems of comparison                                  | 28         |
| 3.3 Problems of prediction                                  | 35         |
| 3.4 Problems of generalization                              | 43         |
| <b>4 Discourse</b>                                          | <b>48</b>  |
| 4.1 Politeness                                              | 49         |
| 4.2 Coherence                                               | 58         |
| 4.3 Discourse transfer and other factors                    | 64         |
| 4.4 Summary and conclusion                                  | 69         |
| <b>5 Semantics</b>                                          | <b>71</b>  |
| 5.1 Propositional semantics                                 | 71         |
| 5.2 Lexical semantics                                       | 77         |
| 5.3 Summary and conclusion                                  | 83         |
| <b>6 Syntax</b>                                             | <b>85</b>  |
| 6.1 Word order                                              | 85         |
| 6.2 Relative clauses                                        | 97         |
| 6.3 Negation                                                | 104        |
| 6.4 Summary and conclusion                                  | 110        |
| <b>7 Phonetics, phonology, and writing systems</b>          | <b>112</b> |
| 7.1 General versus specific predictions                     | 112        |

|           |                                                    |            |
|-----------|----------------------------------------------------|------------|
| 7.2       | Phonetic and phonological transfer                 | 113        |
| 7.3       | Pronunciation, language universals, and typologies | 120        |
| 7.4       | Writing systems                                    | 124        |
| 7.5       | Summary and conclusion                             | 127        |
| <b>8</b>  | <b>Nonstructural factors in transfer</b>           | <b>129</b> |
| 8.1       | Individual variation                               | 130        |
| 8.2       | Transfer and age of acquisition                    | 136        |
| 8.3       | Transfer, linguistic awareness, and social context | 140        |
| <b>9</b>  | <b>Looking back and looking ahead</b>              | <b>151</b> |
| 9.1       | Some caveats                                       | 151        |
| 9.2       | Some conclusions                                   | 152        |
| 9.3       | Some areas for further research                    | 155        |
| <b>10</b> | <b>Implications for teaching</b>                   | <b>157</b> |
|           | Glossary                                           | 165        |
|           | References                                         | 171        |
|           | Language index                                     | 197        |
|           | Author index                                       | 201        |
|           | Subject index                                      | 207        |

## *Series editors' preface*

Language transfer has been a central issue in applied linguistics, second language acquisition, and language teaching for at least a century. Within the last few decades, however, its importance in second language learning has been reassessed several times. In the 1950s it was often deemed the most important factor to consider in theories of second language learning as well as in approaches to second language teaching. In the 1960s its importance waned as learners' errors were seen not as evidence of language transfer but rather of "the creative construction process." Some researchers virtually denied the existence of language transfer in their enthusiasm for universalist explanations. In recent years, however, a more balanced perspective has emerged in which the role of transfer is acknowledged and in which transfer is seen to interact with a host of other factors in ways not yet fully understood.

This reassessment of the significance of language transfer is lucidly demonstrated in this new addition to the Cambridge Applied Linguistics Series. In this timely book, Terry Odlin presents a comprehensive and original account of the nature of language transfer and its role in second language acquisition. Dr. Odlin documents the historical development of the concept of language transfer, explores the role of transfer in discourse, semantics, syntax, phonology, and writing systems, and examines the way language transfer interacts with linguistic as well as cultural, social, and personal factors in second language learning and use. In the process, he surveys a large body of literature and examines data from many different languages.

Dr. Odlin's analysis challenges simplistic notions of language transfer and offers instead a convincing account of the process as a phenomenon that is fundamental to research in second language acquisition and applied linguistics. This book will hence be invaluable to students entering the field of second language acquisition, researchers, language teachers, and anyone interested in the fundamental question of how language systems interact during the process of sec-

ond language acquisition. We are therefore delighted to be able to make Dr. Odlin's research available to a wider audience through the Cambridge Applied Linguistics Series.

Michael H. Long  
Jack C. Richards

# Preface

The significance of cross-linguistic influences has long been a controversial topic. As this book indicates, the controversy has had a long life not only among second language teachers and researchers, but also among linguists interested in questions of language contact and language change. Although it would be too much to hope that this book will cause such a long-standing controversy to die, the discussion of transfer here may help to set to rest some dubious claims and to point the way toward more productive thinking about cross-linguistic influences. While I have tried hard to avoid the sweeping claims that unfortunately have been frequent in discussions of transfer, I make no secret of my belief that transfer is an extremely important factor in second language acquisition. The available evidence, I feel, warrants that belief. Thus, the focus of this book is on empirical investigations of learners' behavior in many contexts. There is some discussion of the pedagogical implications of certain investigations, but it seems to me that relatively little is known about the best ways to make use of transfer research in the classroom – hopefully, more teachers and teacher trainers will begin to think about what those ways are. There is also some discussion of theoretical work in other areas of linguistics, but I have made efforts to limit that discussion, which could go on interminably, and to limit the jargon that usually accompanies such discussion. Readers familiar with *Government and Binding*, *Schema Theory*, and *Sprachbund* will not find those terms, though they will note allusions to research using those terms. Some background in linguistics will be helpful in reading certain chapters (especially Chapter 7), but the glossary provided should help with some of the terminology that seemed impossible to avoid.

While this book has just one author, there are many people who have helped bring about whatever may be praiseworthy in it. In my graduate work I had the good fortune to take courses with Diana Natalicio, who recognized the seriousness of challenges to contrastive analysis in the 1960s and 1970s but who also recognized that the most extreme – albeit fashionable – criticisms of work on transfer were themselves open to challenges. Some of the more novel ideas in this book owe a great deal

to work by Jacquelyn Schachter, Sarah Grey Thomason, and Eric Kellerman, all of whom also provided valuable feedback on a number of my ideas. As this work took shape, Jack Richards provided much encouragement and support – without his interest, this book might never have been finished. Ellen Shaw and Linda Grossman of Cambridge University Press helped in many ways to see the manuscript through the final stages. I would also like to thank several people who made my search for studies of transfer easier by sending me some of their work: Christian Adjemian, David Birdsong, Susan Gass, Lynn Eubank, Markku Filppula, John Hinds, Richard Schmidt, David Singleton, and Lydia White. Many thanks are also due to Lisa Kiser, Alan Brown, and other members of the Department of English at Ohio State who provided valuable comments on earlier drafts of the manuscript. Moreover, I received many forms of assistance from friends and colleagues in the Department of English, the Department of Linguistics, the programs in English as a Second Language, and also from members of the Linguistics Institute of Ireland. In addition, I would like to acknowledge the generous support provided by other units at Ohio State, including the College of Humanities, the Office of Research and Graduate Studies, and the Instructional and Research Computation Center. Finally, I would like to thank my family for their encouragement not only with this project but with much else besides.

# 1 Introduction

When people hear a speaker with a “foreign accent,” they often try to guess the speaker’s background. Sometimes racial features and sometimes a style of clothing will help listeners guess correctly, but often the only reliable clue seems to be how the individual talks. In such cases, questions put to the speaker such as “Are you German?” or “Are you Spanish?” suggest an intuition about the nature of language, an awareness, however unconscious, that the native language of a speaker can somehow cause the individual to sound “foreign” in speaking another language.

The detection of foreign accents is just one example of the awareness that people may often have of *cross-linguistic influence*, which is also known as *language transfer*.<sup>1</sup> That awareness is also evident from time to time in opinions that people have about foreign language study. Many believe that the study of one language (e.g., Latin) will make easier the study of a closely related language (e.g., French). Similarly, people often believe that some languages are “easy” in comparison with others. For example, many English-speaking university students see European languages such as French as less difficult than Oriental languages such as Chinese. Since the similarities between English and French seem to be relatively great, French is often considered “easy.”

An awareness of language transfer is also evident in the mimicking of foreigners. While the representation of foreigners in ethnic jokes is often crude in more ways than one, stereotypes of the way foreigners talk are sometimes highly developed among actors. The following passage comes from a manual to train English-speaking actors in the use of different foreign accents, in this case a Russian one:

Oh! I very good fellow! why? because I Cossack. I very big Cossack. Yah! I captain of Royal Cossack Guard in Moscow – in old country. Oh! I got fifty – hundred – five hundred Cossack they was under me. I be big mans. And womens, they love me lots. Nastia Alexandrovna – she big ballet dancer in Czar ballet – Countess Irina Balushkovna, she love me. All womens they love

1 A more extended definition and also a justification of the term *transfer* appear in Chapter 3 (Section 3.1).

## 2 Language transfer

me. And men? Ach! they be 'fraid from me. They hating me. Why? because I big Cossack. I ride big horse. Drink lots vodka. Oh! I very big mans.  
(Herman and Herman 1943:340)

The manual provides a pronunciation guide for this passage so that actors can make their phonetic mimicry seem plausible, but a number of grammatical features in the passage also seem to be “typically Russian,” such as the absence of an article and a copula in *I very good fellow*. Another passage in the same manual provides a very different linguistic – and ethnic – stereotype. While the Irishwoman’s speech in the following passage might be that of a monolingual speaker of English, it is similar to stereotypical portrayals of Irish-English bilinguals by modern Irish playwrights:

And what business is it of yours that I be awake or no? Be what right do you come snooping after me, following me like a black shadow. Are youse never going to leave me alone? Yous’d be after doing better minding your own business and letting me for to mind mine. For I have an ache in me long-suffering heart and lashin’s of pain cutting through me brain like a dull knife. And me eyes is looking at a world that’s not of your living. For it’s a revelation I’m after having – a view into the banshee world of devils and spirits and the dear departed dead now rotting their whitened bones under the cold, black sod. Ah! sure, now, and it’s the likes of you and your friends that call themselves sane, that disbelieves in what I’m after seeing and knowing. (Herman and Herman 1943:100)

Analogous to the Russian passage, some of the grammatical features in the Irishwoman’s speech appear to be stereotypically Irish: for example, the syntactic pattern in *what I’m after seeing and knowing*, which in standard English would be *what I have seen and known*. While these portrayals of accents may seem exaggerated, they do typify the use of special linguistic structures to characterize the speech of bilinguals.<sup>2</sup>

The distinctiveness of foreign accents often seems understandable in light of cross-linguistic comparisons. For example, Russian does not have present tense copula forms such as *am* or articles such as *a*, and so omissions of the copula and indefinite article in *I very good fellow* may seem to be clearly due to a difference in the grammatical systems of Russian and English. The comparison of such differences, which is known technically as **contrastive analysis**, has long been a part of second language pedagogy, and in the twentieth century contrastive analyses have become more and more detailed.<sup>3</sup> Since such cross-linguistic comparisons constitute an indispensable basis for the study of transfer, the

2 The Irishwoman’s speech is a more accurate characterization than what is often found in so-called Stage Irish (cf. Bliss 1978; Sullivan 1980).

3 Technical terms that appear in the glossary (see page 165) are indicated by boldface at their first occurrence.



discussion of second language research in this book will frequently include contrastive observations.

In light of such everyday abilities as the recognition and mimicry of foreign accents and in light of common beliefs about cross-linguistic similarities and differences, there appears to be a widespread assumption that language transfer is an important characteristic of second language acquisition. It might seem obvious that many characteristics of a learner's linguistic behavior will closely approximate or greatly differ from the actual characteristics of the second language because of similarities and differences predicted by a contrastive analysis. In fact, however, the role of language transfer in second language acquisition has long been a very controversial topic.<sup>4</sup> Some scholars have indeed argued for the importance of transfer; some have gone so far as to consider it the paramount fact of second language acquisition. Yet other scholars have been very skeptical about the importance of transfer. Among linguists and language teachers today, there is still no consensus about the nature or the significance of cross-linguistic influences.

Much of the discussion in the next chapter will review the reasons for the skepticism about transfer, but a brief consideration of one of the most important reasons is appropriate now. As already noted, characteristics of the Russian language seem to explain sentences such as *I very good fellow*. A contrastive explanation, however, seems less than compelling in light of other facts. For example, speakers of Spanish, which, like English, has copula verb forms, frequently omit forms such as *am* and *is* (cf. Section 2.2). Moreover, such errors are found not only among Russian and Spanish speakers but also among speakers of other languages – and also among children learning English as their native language. Thus, while a contrastive analysis might explain a Russian speaker's omission of copula forms, a Spanish-English contrastive analysis would not explain the same error, and a contrastive analysis is irrelevant for monolingual children who make this same error as they acquire English. The pervasiveness of certain types of errors has thus been among the most significant counterarguments against the importance of transfer.

Despite the counterarguments, however, there is a large and growing

4 The terms *acquisition* and *learning* will be used interchangeably throughout this work even though much of the writing on second language acquisition (e.g., Krashen 1981) distinguishes between the two terms. I agree with Krashen and others that the outcomes of acquisition can differ depending on the awareness of language that individuals have (cf. Section 8.3). However, I strongly disagree with Krashen's analysis of transfer and with much else in his interpretation of second language acquisition (cf. Sections 2.2, 3.1). Since his characterization of *acquisition* and *learning* is questionable in several respects, I see no reason to use his terminological distinctions (cf. Gregg 1984; Odlin 1986).

body of research that indicates that transfer is indeed a very important factor in second language acquisition. Accordingly, the primary aim of this book is to reconsider the problem of transfer in light of recent second language research. While the research to be reviewed points to the importance of transfer, it also frequently points to the importance of other significant factors in second language acquisition. Thus, even though a comprehensive review of second language research is beyond the scope of this book, there will be frequent discussion of cases in which transfer is either not a significant influence or an influence that interacts with other influences.

There are a number of reasons for language teachers and linguists to consider more closely the problem of transfer. Teaching may become more effective through a consideration of differences between languages and between cultures. An English teacher aware of Spanish-based and Korean-based transfer errors, for example, will be able to pinpoint problems of Spanish-speaking and Korean-speaking ESL students better, and in the process, communicate the very important message to students that their linguistic and cultural background *is* important to the teacher.<sup>5</sup> Also, consideration of the research showing similarities in errors made by learners of different backgrounds will help teachers to see better what may be difficult or easy for anyone learning the language they are teaching.

There are yet other reasons to know about research on transfer. For historical linguists, such knowledge can lead to insights about the relation between language contact and language change. Although languages change for a variety of reasons, the bilingualism that often results from language contact situations can be a major factor. For example, Hiberno-English, the dialect spoken in parts of rural Ireland, does have several of the unusual characteristics of the Irishwoman's speech cited earlier, and a number of those characteristics appear to result from the influence of Irish. Research on transfer is also important for a better understanding of the nature of language acquisition in any context and is thus of interest to anyone curious about what is common to all languages, that is, *language universals*. As Comrie (1984) has noted, second language research can provide a valuable empirical check on the merit of universalist theories, and the issue of transfer is likely to figure prominently in such research.

This book consists of ten chapters. The next two provide an overview of the issues: Chapter 2 is a historical survey of the controversy sur-

5 Throughout this book, the term *ESL* (English as a Second Language) will be used even in cases in which *EFL* (English as a Foreign Language) might be more appropriate. While such a terminological distinction can be crucial for those developing syllabi or preparing pedagogical materials, the distinction is less important for researchers studying cross-linguistic influence.

rounding language transfer, and Chapter 3 is a discussion of four types of problems especially important in the investigation of transfer. The next four chapters survey second language research on transfer and universals in relation to linguistic subsystems: discourse (Chapter 4); semantics – including a discussion of morphology (Chapter 5); syntax (Chapter 6); and phonetics, phonology, and writing systems (Chapter 7). Chapter 8 discusses in more detail some aspects of language transfer which structural descriptions cannot always account for, such as the effects of individual variation in second language acquisition. Chapter 9 reviews important currents in the research discussed in earlier chapters, and Chapter 10 considers some of the implications that the research may have for teaching.

## **Further reading**

Most studies of transfer appear in a wide variety of journals, but they sometimes appear in special collections. One of the best collections is edited by Gass and Selinker (1983). A recent book-length study by Ringbom (1987) combines a review of many of the controversies about transfer with a detailed empirical study. Ellis (1985) has written a remarkably comprehensive and judicious survey of research on second language acquisition, including work on transfer. For more discussion of linguistic analyses of the literary treatment of foreign accents, a text by Traugott and Pratt (1980) is useful. Recent introductions to linguistics include texts by Bolinger and Sears (1981) and Fromkin and Rodman (1983).

## 2 *Earlier thinking on transfer*

Discussions of transfer often begin with the work of American linguists in the 1940s and 1950s. Yet while the work of Charles Fries, Robert Lado, and others was clearly a major catalyst for subsequent research, serious thinking about cross-linguistic influences dates back to a controversy in historical linguistics in the nineteenth century. Accordingly, this chapter begins with a discussion of that controversy among scholars whose primary interests were not second language acquisition or language teaching but rather language classification and language change. The controversy promoted work on language contact that overlaps considerably with more recent studies of second language acquisition. Because the thinking of Fries, Lado, and others prompted much of the growth of research in second language acquisition, their views receive considerable attention, as do the views of some who have been very critical of their work. While this chapter can give only a suggestion of the historical context of the polemics on transfer, it provides important background for some fundamental issues discussed in subsequent chapters.

### 2.1 Languages (and dialects) in contact

#### *Historical change and language mixing*

Language contact situations arise whenever there is a meeting of speakers who do not all share the same language and who need to communicate.<sup>1</sup> When the communicative needs of people go beyond what gestures and other paralinguistic signals can achieve, some use of a second language becomes necessary. The languages learned in contact situations may or may not show some kind of **language mixing**, that is, the merging of characteristics of two or more languages in any verbal communication.

1 There are other kinds of language contact besides those discussed in this chapter, as, for example, when a French scholar deciphers a text in ancient Egyptian. Such cases, though, are exceptional in a number of ways.

If mixing does occur, native language influence is only one of the possible forms it can take. Another kind of mixing is in the form of *borrowings* from a second language into the native language (e.g., the use by English speakers of the loanword *croissant* from French to describe a certain kind of pastry), and still another kind is **code-switching**, in which there is a systematic interchange of words, phrases, and sentences of two or more languages (cf. Sections 8.2, 8.3).

People often show some awareness of language mixing, even though they are usually not familiar with terms such as *code-switching* and *transfer* (cf. Chapter 1 and Section 8.3). Among Indians in the Vaupés region of the Amazon rain forest, for example, there is a keen awareness of the mixing that arises in their multilingual villages (Sorenson 1967; Jackson 1974). Such awareness probably reflects a consciousness going far back into prehistory. Whether in the rain forest or elsewhere, humans have often seen themselves as belonging to different social groups, and they have often considered language to be an important distinguisher of their own group from others; it is no accident that the names of languages frequently designate ethnic groups (e.g., *Chinese*, *Navajo*, and *English*). Accordingly, any introduction of loanwords or other kinds of language mixing may be viewed either as a kind of linguistic intrusion or as a “foreign import,” sometimes welcome, sometimes not (cf. Le Page and Tabouret Keller 1985). It is significant that the Indians of the Vaupés do not look upon language mixing favorably; although that attitude is probably not universal, people in many communities do have similarly unfavorable attitudes toward various kinds of mixing. For example, English loanwords in French and other languages have frequently been a target of language purists (cf. Section 8.3).

From antiquity onwards there is a historical record of people associating language contact and mixing with “contamination” (Silvestri 1977; Thomason 1981). Typical of such associations were scholarly discussions in Renaissance Europe about the link between Latin and the vernacular languages related to it. With regard to the origins of French, for example, scholars speculated about how speakers of other languages may have “corrupted” the language brought to Gaul by the Romans (Silvestri 1977). Although some scholarly work before the nineteenth century did make specific claims, most of the discussion about language contact and mixing was rather nebulous. Apart from occasional remarks about loanwords, few discussions included either detailed characterizations of the nature of cross-linguistic influences or specific examples of such influences (Silvestri 1977).

In the nineteenth century, debate about the importance of language contact and mixing intensified. The question of mixing had major implications for two interrelated problems that interested many nineteenth-century linguists: language classification and language change. The

steady accumulation of grammars of languages in every part of the world made ever clearer the diversity of human languages and the scientific challenge of classifying them (Robins 1979). Many scholars came to believe that grammar was the soundest basis on which to construct classifications. Aware that lexical borrowings (i.e., loanwords) could often make classification decisions difficult, scholars often expected to find in grammar a linguistic subsystem unaffected by language contact and thus a key to distinguish any language. Müller (1861/1965:75), for example, was well aware of the large number of loanwords from Latin, French, and other languages in English, but considered English grammar to be immune from cross-linguistic influences:

The grammar, the blood and soul of the language, is as pure and unmixed in English as spoken in the British Isles, as it was when spoken on the shores of the German ocean by the Angles, Saxons, and Juts [sic] of the continent.<sup>2</sup>

The beliefs of Müller and others about the uniqueness of grammar were usually related to assumptions about the **tree model** of language change in which languages are viewed as parts of a “family tree.” In that model, Latin, for example, is characterized as the “parent” language and French, Spanish, Rumanian, and other Romance tongues as the “daughter” languages.<sup>3</sup> The pattern of change in the tree model is primarily one of *internal development*, in which characteristics of the parent language undergo changes that are systematically manifested in the daughter languages. For example, in French, Spanish, and Rumanian, noun phrases commonly have definite articles, as in the following translations of “the mountain”:

French: *le mont*

Spanish: *el monte*

Rumanian: *munte-le*

In Latin there were no articles, but in virtually all of the daughter languages there are definite and indefinite articles. The development of such an innovation in each of the daughter languages suggests the existence of what Whitney (in a slightly different connection) called “forces which are slowly and almost insensibly determining the growth of a language” (1881:25). Like Müller, Whitney took a dim view of the notion of mixing, and their attitude was shared by other scholars aware of the massive evidence of internal change not only in morphology and syntax but also in phonetics and phonology (e.g., Meillet 1948). If internal

2 Despite the racist connotations of this passage, Müller was an outstanding advocate of racial understanding in the nineteenth century.

3 As Robins (1979) observes, such characterizations of historical relationships are metaphoric and only partially revealing. Nevertheless, the family tree metaphor had an enormous impact on thinking about language change.

development could explain so much about language change, were explanations involving language mixture really necessary?

### *Areal linguistics*

In the judgment of many scholars in the nineteenth and twentieth centuries, the answer to this question is affirmative, despite the cogent arguments by Whitney and others. Even while the evidence supporting the tree model of change is strong, there is good evidence supporting a very different model, the *wave model* (cf. Bloomfield 1933; Bailey 1973; Bynon 1977; Zobl 1984). First developed in the nineteenth century, the wave model has long been recognized as a useful complement to the tree model, especially for an understanding of *dialect change*. While characterizations of the wave model have changed over the years, the model still posits that linguistic patterns in one dialect can affect another dialect considerably, especially if the dialects are spoken in adjoining regions. For example, the English spoken in Chicago and that spoken in nearby towns in northern Illinois are similar, but recent changes in pronunciation that have appeared in Chicago make that variety different from other Illinois dialects. The changes seem to be slowly spreading from Chicago into other Illinois towns, largely through contacts between Chicagoans and people in the larger towns (cf. Callary 1975; Chambers and Trudgill 1980).

There is now a considerable body of scholarship pointing to the importance of dialect mixture (e.g., Trudgill 1986). Yet the significance of the wave model is not limited to dialect contact. As most linguists acknowledge, the difference between languages and dialects is often fuzzy, as the linguistic situation in parts of Spanish-speaking Uruguay and Portuguese-speaking Brazil shows. While Spanish and Portuguese are distinct in many ways, they might well be considered two Romance dialects instead of two languages were it not for political facts. On the border between Brazil and Uruguay there have been frequent contacts between people of both nations, and, not surprisingly, the similarity of Spanish and Portuguese has encouraged a great deal of mixing which one might call either dialect mixing or language mixing (Rona 1965).

Even when the differences between two languages are greater than is the case with Spanish and Portuguese, there is a possibility of language mixing. For example, Rumanian and Bulgarian have somewhat different "genetic" classifications (the former is a Romance language and the latter a Slavic language), but centuries of contact between speakers of Rumanian, Bulgarian, and other languages in the Balkans have led to many *areal* (i.e., regional) similarities not due to internal changes (Sandfeld 1930; Joseph 1983a). The definite article, for instance, follows the noun